Curtin University
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Network System Design

LOGICAL NETWORK ARCHITECTURE DESIGN FOR GREENFIELD UNIVERSITY

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1. Introduction

1.1 Introduction

This document presents the logical network architecture for the Greenfield University multi-campus network redesign. This Stage 2 design is based on the detailed requirements analysis from Stage 1 (Assignment 1) and maps out the identified business, technical, security and performance requirements into a scalable, secure and high performance logical infrastructure. The design resolves the key issues that were identified in the current 12-year-old flat network, such as insufficient bandwidth, no VLAN segmentation, no QoS, single points of failure, and inadequate security architecture.

Greenfield University has three campuses in a metropolitan area: Campus A (main campus) which houses most of the academic departments, administrative offices and central data centre; and two satellite campuses, Campus B and Campus C, which offer additional teaching facilities, research laboratories and student accommodation. The university has about 18,000 students and 2,400 staff spread across 10 main buildings, with a peak aggregate demand of 10-12 Gbps during peak academic hours (10:00 AM – 2:00 PM) when students are accessing the LMS, video conferencing and cloud services concurrently.

The current network has been plagued by connectivity issues, security concerns, bandwidth limitations, and scalability issues. The flat architecture with a single core switch, unmanaged 100 Mbps access switches, static routing, no VLANs, no QoS, and no centralised monitoring has caused many service interruptions, especially during peak hours. Sensitive data is stored on a single server room without a DMZ, putting it at unnecessary risk on mission critical servers such as LMS (Canvas), student records, financial systems, and email.

1.2 Project Overview

The network redesign project is designed to develop a comprehensive, scalable and secure logical network design that resolves all critical issues identified in Stage 1. The project has five sets of objectives:

1.2.1 Business Goals

- Ensure reliable and high-speed connectivity throughout all campuses to facilitate the university's digital transformation.
- Minimise TCO by consolidating network management and services into the Campus A data centre.
- Facilitate easy collaboration between different campuses for students and staff, regardless of their location.
- Improve the competitiveness of the university by creating a learning environment that is rich in technology.
- Support hybrid and remote learning and provide equitable access to learning off campus.

1.2.2 Technical Goals

- Create a logical traffic separation and troubleshooting friendly hierarchical core-distribution-access architecture.
- Use VLAN segmentation to separate student, staff, guest, research and administrative traffic.
- Redundant links between core and distribution layers with failover less than 5 seconds.
- Set up a centralised data centre with redundant 10 Gbps fibre connections to all campuses.
- Use VLSM to implement a structured IP addressing scheme to reduce wasted address space and to enable route summarisation.

1.2.3 Security Goals

- Use VLANs and firewall policies to isolate sensitive administrative and financial systems.

- Implement next generation firewalls at the Internet edge and between the data centre and campus networks.
- Design a DMZ segment for the public facing services (university website, LMS gateway, email relay).
- Use IPsec VPN tunnels to encrypt all traffic between campus to prevent eavesdropping.
- Use 802.1X port-based authentication for all wired and wireless access ports.

1.2.4 Scalability Goals

- Support 20% annual growth for 3 years to about 24,600 users without redesigning the core.
- Up to two new campuses can be added within 5 years with VLAN trunking and IP extension capability.
- Plan for future departmental growth by reserving IP addressing blocks per campus and per building.
- Identify modular switch chassis for core and distribution layers for incremental line card and uplink additions.
- Support new sites with dynamic route propagation for routing protocol selection.

1.2.5 Performance Goals

- Provide end-to-end latency of less than 50 milliseconds between any campus and the central data centre.
- Connect 10 Gbps distribution-to-core uplinks to each access switch, providing minimum 1 Gbps to each access switch.
- Ensure 99.99% uptime for mission critical systems (less than 52 minutes of unplanned downtime per year).
- Ensure that less than 1% packet loss for high priority traffic classes during peak hours.
- Support 5,000 concurrent video conferencing sessions with less than 100ms latency and less than 30ms jitter.

1.3 Design Objectives

The following specific objectives are derived from the requirements identified in Stage 1 and are the goals of the logical network design presented in this document:

Design Objective	Stage 1 Requirement	Design Solution
Eliminate flat network architecture	Challenge C2: No network segmentation or VLANs	Hierarchical core-distribution-access model with VLAN segmentation per user group
Upgrade inter-campus bandwidth	Challenge C1: Insufficient 1 Gbps links	Redundant 10 Gbps fibre links between all campuses
Remove single points of failure	Challenge C4: No redundancy	Dual core switches, dual distribution uplinks, dual ISP connectivity
Implement traffic prioritisation	Challenge C3: Absence of QoS	DiffServ with DSCP marking, LLQ for real-time traffic, WRED for congestion avoidance
Establish security zones	Challenge C5: No DMZ or adequate security	DMZ for public services, isolated Finance VLAN, firewall-enforced security zones
Enable dynamic failover	Challenge C6: Static routing only	OSPF for internal routing with sub-second convergence; BGP for internet edge
Implement centralised monitoring	Challenge C7: No network management	SNMP, NetFlow, and IP SLA framework for

		proactive fault detection and capacity planning
Modernise access layer	Challenge C8: 100 Mbps unmanaged switches	1 Gbps managed access switches with PoE support and VLAN trunking
Deploy network access control	Challenge C9: No 802.1X	802.1X port-based authentication with RADIUS and restricted legacy VLAN
Encrypt inter-campus traffic	Challenge C10: No encrypted communication	IPsec VPN tunnels over all inter-campus WAN links

Table 1: Design Objective Alignment with Stage 1 Challenges and Proposed Solutions

1.4 Summary of the Requirements for Stage 1

In Stage 1, ten critical challenges, eleven validated assumptions, seven project constraints and ten risk considerations were identified. The ten challenges that directly inform this logical design are summarised in Table 2:

Challenge ID	Description	Impact on Design
C1	Insufficient bandwidth between campuses (single 1 Gbps links)	Redundant 10 Gbps inter-campus links with OSPF and ECMP
C2	No network segmentation or VLAN architecture	VLAN segmentation isolating six user/security zones per campus

C3	Absence of QoS and traffic prioritisation	DiffServ/DSCP with six traffic classes and WRED
C4	Single points of failure throughout the network	Dual core switches, dual links, dual ISP, <5s failover
C5	No DMZ or adequate security architecture	DMZ segment, next-gen firewalls, isolated Finance VLAN
C6	Static routing with no dynamic failover	OSPF for internal routing, BGP for internet edge
C7	No centralised network monitoring or management	SNMP v3, NetFlow v9, IP SLA for proactive monitoring
C8	Outdated access layer (100 Mbps unmanaged switches)	1 Gbps managed PoE switches with VLAN trunking
C9	No 802.1X or network access control	802.1X with RADIUS, restricted legacy VLAN for non-compliant devices
C10	No encrypted inter-campus communication	IPsec VPN tunnels over all WAN links

Table 2: Stage 1 Critical Challenges and Their Impact on Logical Design

The stakeholder requirements identified in Stage 1 are: 18,000 students need reliable access to LMS and video conferencing; 2,000 academic staff require high bandwidth

research data transfer and low latency teaching tools; 400 administrative staff require secure access to the financial system with zero tolerance; and 100 senior management require strategic visibility and regulatory compliance. The hierarchical architecture, VLAN segmentation, QoS prioritisation and security zones described in the following sections address these requirements.

1.5 Design Assumptions

The validated assumptions of Stage 1 are used as the basis for the logical network design, with the design-specific assumptions of Stage 2 added to complete the design:

Assumption ID	Assumption	Justification
A1	Approximately 20,500 total users (18,000 students, 2,400 staff) with 20% annual growth reaching ~24,600 users	Based on enrolment projections; addressing scheme sized to accommodate growth without re-architecture
A2	Campuses connected by existing fibre capable of 10 Gbps with diverse overhead fibre for redundancy	Typical metropolitan university infrastructure; supports redundant 10 Gbps inter-campus backbone
A3	Central data centre has sufficient power, cooling, and space for upgrades	Existing 12-year operational data centre with expansion capacity
A4	All end-user devices support 802.1X authentication and DHCP	Modern OS (Windows 10+, macOS, Linux) natively support 802.1X
A5	Peak load 10:00 AM – 2:00 PM and 6:00 PM – 9:00 PM for LMS	Traffic load analysis and typical university usage patterns

A6	Two ISPs provide combined ~5 Gbps with BGP multihoming support	Standard enterprise-grade ISP connections in metropolitan area
A7	Research transfers scheduled 11:00 PM – 7:00 AM to reduce contention	Long-duration batch jobs typical in research; aligns with traffic load analysis
A8	Existing fibre in serviceable condition requiring only transceiver upgrade	Consistent with infrastructure reuse constraint; avoids new fibre installation cost
A9	Cloud services accessed via public internet, not dedicated peering	Standard university model; direct peering excluded from scope
A10	Legacy devices without 802.1X placed on restricted VLAN	Practical assumption for 12-year-old network equipment
A11 (New)	Cisco 3560/3650 and 2960 switches used as logical equivalents for design validation	Packet Tracer device limitations; actual procurement to follow vendor evaluation
A12 (New)	IPv6 deployment uses Unique Local Addresses (ULA) during dual-stack transition	RFC 4193 ULA blocks (fd00::/48) provide internal addressing without global allocation

Table 3: Validated Design Assumptions from Stage 1 and Stage 2

2. Logical Network Design

This section presents the logical network architecture for Greenfield University's multi-campus network redesign. The logical diagram provides a high-level visual representation of the hierarchical network topology, VLAN segmentation, device interconnections, and traffic flows across all three campuses. The diagram serves as a blueprint for the detailed addressing, routing, performance, and security architectures described in subsequent sections.

2.1 Network Topology Overview

The Greenfield University network follows a hierarchical three-tier core-distribution-access architecture, which is the industry-standard design for large-scale enterprise and campus networks [1]. This model was selected because it provides logical separation of traffic, easier troubleshooting, and independent scaling of each network layer [1].

The topology consists of three campuses connected through a hybrid star-mesh design:

Campus	Type	User Count	Role in Topology
Campus A	Main Campus	11,520 users	Core hub with central data centre
Campus B	Satellite Campus	5,620 users	Spoke connected via redundant 10 Gbps links
Campus C	Satellite Campus	3,360 users	Spoke connected via redundant 10 Gbps links

Table 4: Hybrid Star-Mesh Topology

Topology Justification:

- Star topology is used for inter-campus connectivity, with Campus A as the central hub. This design minimises latency between satellite campuses and the central data centre, where most mission-critical servers (LMS, financial systems, student records) are hosted [1].

- Full-mesh connectivity is implemented at the core layer within Campus A, with dual core switches providing redundancy and load sharing [1].
- Partial-mesh connectivity is used at the distribution layer, where each distribution switch connects to both core switches for redundancy [1].

2.2 VLAN and Subnet Segmentation

The network is logically segmented into multiple VLANs based on user roles, device types, and security requirements. This segmentation provides security isolation, broadcast domain control, and simplified policy enforcement [1].

2.2.1 Campus A VLAN Allocation

VLAN ID	VLAN Name	Subnet	Mask	Gateway	Max Hosts	Purpose
10	Students-A	10.1.10.0	/22	10.1.10.1	1,022	Student data and devices
20	Staff-A	10.1.20.0	/23	10.1.20.1	510	Academic and administrative staff
30	Admin-A	10.1.30.0	/24	10.1.30.1	254	Administrative and executive services
40	Research-A	10.1.40.0	/23	10.1.40.1	510	Research laboratories and HPC access
50	Guest-A	10.1.50.0	/23	10.1.50.1	510	Visitor and recreational WiFi
60	CCTV-A	10.1.60.0	/24	10.1.60.1	254	IP surveillance cameras

80	Mgmt-A	10.1.80.0	/24	10.1.80.1	254	Switch and router management
90	Native-A	10.1.90.0	/24	—	—	Native trunk VLAN (unused data)
100	DMZ-A	10.1.100.0	/24	10.1.100.1	254	Public-facing services (website, LMS gateway)
110	Servers-Int-A	10.1.110.0	/24	10.1.110.1	254	Internal servers (LMS, DNS, DHCP, records)
120	Finance-A	10.1.120.0	/24	10.1.120.1	254	Financial systems (isolated, zero-tolerance)

Table 5: Campus A VLAN Allocation

2.2.2 Campus B and C VLAN Allocation Templates

VLAN ID	VLAN Name	Campus B Subnet	Campus C Subnet	Purpose
10	Students	10.2.10.0/22	10.3.10.0/22	Student data and devices
20	Staff	10.2.20.0/23	10.3.20.0/23	Academic and administrative staff
40	Research	10.2.40.0/23	10.3.40.0/23	Research laboratories

50	Guest	10.2.50.0/23	10.3.50.0/23	Visitor and recreational WiFi
80	Mgmt	10.2.80.0/24	10.3.80.0/24	Switch and router management
90	Native	10.2.90.0/24	10.3.90.0/24	Native trunk VLAN

Table 6: Campus B and C VLAN Allocation

2.2.3 Point-to-Point Link Allocation

Inter-device links use /30 subnets to conserve address space [1]:

Link Purpose	Subnet	Device A IP	Device B IP
Core-A1 ↔ Dist-A-Acad (Primary)	10.1.250.0/30	10.1.250.1	10.1.250.2
Core-A2 ↔ Dist-A-Acad (Backup)	10.1.250.16/30	10.1.250.17	10.1.250.18
Core-A1 ↔ Dist-A-Admin (Primary)	10.1.250.4/30	10.1.250.5	10.1.250.6
Core-A2 ↔ Dist-A-Admin (Backup)	10.1.250.20/30	10.1.250.21	10.1.250.22
Core-A1 ↔ R-A-WAN	10.1.250.32/30	10.1.250.33	10.1.250.34
Core-A2 ↔ R-A-WAN	10.1.250.36/30	10.1.250.37	10.1.250.38
Core-A1 ↔ FW-A-Edge	10.1.250.40/30	10.1.250.41	10.1.250.42
Core-A2 ↔ FW-A-Edge	10.1.250.44/30	10.1.250.45	10.1.250.46

Table 7: Link Allocation

2.3 Key Network Devices and Interconnections

The logical network diagram includes the following key devices organised by layer:

2.3.1 Core Layer (Campus A)

Device	Quantity	Role	Redundancy
Core Switch A1	1	Primary core switch (Active)	Yes (paired with Core A2)
Core Switch A2	1	Secondary core switch (Standby)	Yes (paired with Core A1)
WAN Router A	1	Inter-campus routing	Yes (dual links to B and C
Edge Firewall Cluster	2	Internet edge security	Active-Active cluster

Table 8: Core Layer

Core Layer Functions:

- High-speed packet switching backbone (100 Gbps)
- Interconnection between Campus A distribution switches
- Connectivity to satellite campuses via WAN routers
- Internet connectivity through edge firewalls [1]

2.3.2 Distribution Layer

Device	Location	Quantity	Role
Dist-A-Acad	Campus A - Academic Buildings	2	Aggregates traffic from academic access switches
Dist-A-Admin	Campus A - Admin Building	2	Aggregates administrative access switches
Dist-A-DC	Campus A - Data Centre	2	Aggregates server traffic

Dist-A-Sec	Campus A - Security Centre	2	Aggregates CCTV and access control
Dist-B-Core	Campus B	2	Core/distribution for satellite campus
Dist-C-Core	Campus C	2	Core/distribution for satellite campus

Table 9: Distribution Layer

Distribution Layer Functions:

- Inter-VLAN routing (Layer 3 switching)
- QoS policy enforcement (queuing and scheduling)
- Route summarisation towards core
- Broadcast domain demarcation [1]

2.3.3 Access Layer

Device Type	Quantity	Location	Features
Access Switches	120 total	Across all campuses	1 Gbps PoE+, VLAN trunking, 802.1X
Wireless APs	450	Lecture halls, libraries, common areas	Wi-Fi 6 (802.11ax)
Wireless Controllers	2	Campus A Data Centre	Centralised AP management

Table 10: Access Layer

Access Layer Functions:

- Direct connectivity to end-user devices
- VLAN assignment based on user role (802.1X)
- DSCP marking for QoS
- PoE+ power for IP phones and access points [1]

2.3.4 Inter-Campus Connectivity

Connection	Bandwidth	Redundancy	Security
Campus A ↔ Campus B	10 Gbps	Dual diverse fibre paths	IPsec VPN tunnel
Campus A ↔ Campus C	10 Gbps	Dual diverse fibre paths	IPsec VPN tunnel
Campus A ↔ ISP 1	5 Gbps	Dual ISP	BGP + NGFW
Campus A ↔ ISP 2	5 Gbps (failover)	Dual ISP	BGP + NGFW

Table 11: Inter-Campus Connectivity

2.4 Design Rationale

2.4.1 Alignment with Stage 1 Requirements

The logical network topology directly addresses all ten challenges identified in Stage 1:

Challenge	How the Topology Addresses It
C1: 1 Gbps inter-campus links	Redundant 10 Gbps links between all campuses
C2: No VLAN segmentation	11 VLANs per campus isolating user groups
C3: No QoS	QoS policies at access/distribution/core layers
C4: Single points of failure	Dual core switches, redundant links, dual ISP
C5: No DMZ	Dedicated DMZ VLAN (100) with firewall isolation
C6: Static routing	OSPF dynamic routing with sub-second convergence
C7: No monitoring	SNMP, NetFlow, IP SLA integrated into topology
C8: 100 Mbps access switches	1 Gbps managed PoE switches (120 total)
C9: No 802.1X	802.1X authentication at all access ports

C10: No encryption	IPsec VPN tunnels on all inter-campus links

2.4.2 Why Three-Tier Architecture?

The three-tier core-distribution-access model was selected because:

1. Scalability: Each layer can be expanded independently without redesigning the entire network. Access switches can be added for new buildings, distribution switches can be added for new functional blocks, and core switches can be upgraded for higher capacity [1].
2. Modularity: The modular approach enables "pay-as-you-grow" expansion. Each building is served by a separate distribution layer, providing flexibility in adding new buildings with minimal impact on existing ones [1].
3. Fault Isolation: Issues at one layer are contained within that layer. For example, a broadcast storm on an access switch does not affect the distribution or core layers.
4. Traffic Optimization: Traffic between devices on the same access switch stays local; traffic between buildings is routed at the distribution layer; and inter-campus traffic traverses the core layer. This minimises unnecessary traffic on higher-capacity links [1].

2.4.3 Why Hybrid Star-Mesh Topology?

- Star topology for inter-campus connectivity minimises latency between satellite campuses and the central data centre, where most mission-critical servers are hosted. A star topology also simplifies routing and network management [1].
- Full-mesh topology at the core layer provides maximum redundancy and load sharing between the dual core switches. Both switches are active, sharing traffic via ECMP [1].
- Partial-mesh topology at the distribution layer provides redundancy without excessive complexity. Each distribution switch connects to both core switches, ensuring automatic failover if one core switch or link fails [1].

2.4.4 Addressing Hierarchy

The addressing scheme follows a four-level hierarchy aligned with the physical campus topology:

- Level 1 – Campus Boundary (/16): The first two octets (10.x) define the campus. This allows route summarisation at the WAN edge, with just one OSPF area or BGP advertisement per campus.
- Level 2 – Functional Zone (/22 to /24): The third octet identifies devices by function (students, staff, servers, management). This logical grouping simplifies ACLs, firewall rules, and QoS policies.
- Level 3 – Building/Floor (Host ID): The fourth octet is used for individual devices. Each Student /22 subnet is divided into four /24 blocks, representing different buildings or floors.
- Level 4 – Point-to-Point Links (/30): Inter-device links are dedicated /30 subnets to reduce waste, with 62 spare /30 subnets in the 10.x.250.0/24 block for future expansion.

2.4.5 IPv6 Dual-Stack Justification

The topology supports IPv6 dual-stack operation, which enables the university to maintain IPv4 services while gradually migrating to IPv6 [1]. Each campus is allocated a /48 Unique Local Address (ULA) block (fd00:x::/48) per RFC 4193, which is further divided into /64 VLAN subnets, one-to-one corresponding to the existing IPv4 VLANs.

Benefits of this approach:

- Backward compatibility: Existing IPv4-only devices continue to function.
- Future-proofing: The network is ready for IPv6-only operation when required.
- Simplified management: One-to-one mapping between IPv4 VLANs and IPv6 /64s makes troubleshooting easier.

2.5 Logical Network Diagram

The logical network diagram is provided separately as a visual representation of the architecture described above. The diagram includes:

1. **All three campuses** (Campus A, B, and C) with user counts
2. **Hierarchical layers** (Core, Distribution, Access) clearly labelled
3. **VLAN segmentation** per campus
4. **Key network devices** (routers, switches, firewalls, WLAN controllers)
5. **Inter-campus links** with bandwidth labels (10 Gbps Redundant)
6. **Internet edge** with dual ISP connections
7. **Data Centre** with server clusters
8. **DMZ** segment
9. **Traffic flow arrows** showing direction
10. **Legend** with colour coding and line types

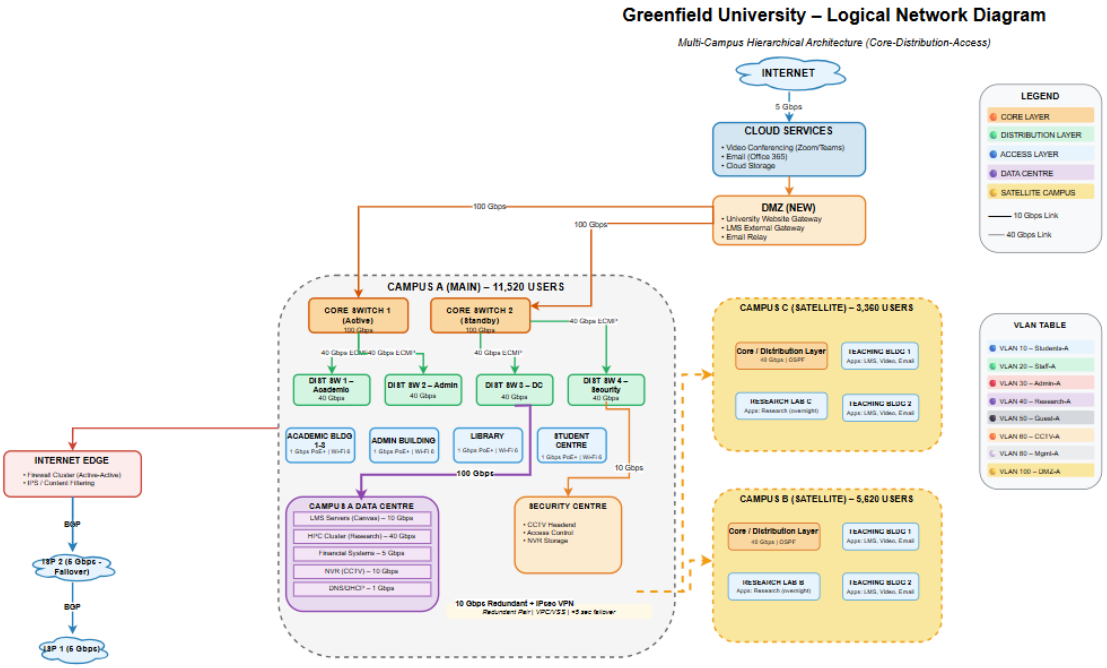


Figure 1: Logical Network Diagram

3. Network Addressing Architecture

This section describes the structured IP addressing scheme for the Greenfield University multi-campus network. The architecture uses Variable Length Subnet Masking (VLSM) in the private IPv4 space 10.0.0.0/8 and allocates addresses in a hierarchical manner to achieve maximum routing efficiency, minimize address waste, and accommodate future growth. A dual-stack IPv6 approach with Unique Local Addresses (ULA) provides long-term scalability and backwards compatibility during the transition period.

3.1 IPv4 and IPv6 Addressing Plan

The addressing plan provides a /16 per campus, which provides a clear hierarchical structure that enables route summarisation at the WAN edge and makes troubleshooting easier. Table 4 shows the overall allocation:

Campus	IPv4 Block	IPv6 ULA Block	Purpose
Campus A (Main)	10.1.0.0/16	fd00:1::/48	Core campus, data centre, 11,520 users
Campus B (Satellite)	10.2.0.0/16	fd00:2::/48	Teaching and accommodation, 5,620 users
Campus C (Satellite)	10.3.0.0/16	fd00:3::/48	Teaching and research, 3,360 users
Inter-Campus WAN	10.255.0.0/16	fd00:ff::/48	Point-to-point links and router loopbacks
Reserved Future 1	10.4.0.0/16	fd00:4::/48	Expansion campus D
Reserved Future 2	10.5.0.0/16	fd00:5::/48	Expansion campus E

Table 12: Hierarchical IPv4 and IPv6 Address Allocation per Campus

The /16 allocation per campus gives 65,534 usable host addresses, which is enough to support the current user population and has ample room for the expected 20% growth per year over the next three years. Summarisation boundaries match campus boundaries: One route advertisement (e.g., 10.1.0.0/16 for Campus A) decreases the size of the routing table at the WAN edge and enhances convergence when topology changes.

IPv6 has built-in security features, including the mandatory support for IPsec as described in Tutorial 06 (Section 6.6). IPsec is required for IPv6-capable devices, but is not enabled by default, it must be enabled. The university will use IPsec in transport mode for traffic between campuses and tunnel mode for remote access VPNs. Extension headers (Authentication Header and Encapsulating Security Payload) offer integrity, authentication and confidentiality, but do not suffer from NAT traversal problems as seen in IPv4-only environments. This is a native security capability that is a major benefit of the dual-stack migration approach.

During the transition period, the dual-stack method is used for IPv6, which is recommended in Tutorial 06: Enterprise IPv6 migration (Tian and Gao, 2024). Each campus is allocated a /48 Unique Local Address (ULA) block (fd00:x::/48) per RFC 4193, which is further divided into /64 VLAN subnets, one-to-one corresponding to the existing IPv4 VLANs. This allows network administrators to use either address family for the same logical segment, making troubleshooting and policy enforcement easier. While the university is migrating to IPv6, the internal routing and services will be based on ULA, and Global Unicast Addresses (GUA) will be assigned when the university has an IPv6 peering relationship with its ISP. NAT64/DNS64 mechanisms are kept as future options for legacy systems that are still using IPv4.

During the transition period, IPv6-over-IPv4 tunneling (RFC 4213) ensures IPv6 packets can traverse IPv4-only infrastructure segments, such as legacy ISP links or third-party research partner networks. An IPv6 datagram is encapsulated in an IPv4 header with the Protocol field set to 41 (IPv6), so that it can pass through IPv4 routers without being dropped. Where dual-stack is not yet deployed, the university will employ configured tunnels (e.g., 6in4) for inter-campus links, and will terminate the tunnels at the campus

edge routers. This mechanism guarantees that there is always IPv6 service during the phased migration.

3.2 VLAN and Subnet Allocation

VLSM is used within each campus /16 to allocate subnets to accommodate host density and security needs. The Campus A allocation is shown in Table 5, and is a copy of this allocation with campus-specific address prefixes for Campuses B and C.

The VLAN and subnet allocation is based on the variable-length subnet masking (VLSM) method recommended by Tutorial 06 (Tian and Gao, 2024). The four-step design process allows for efficient use of addresses and allows for future expansion. This is in contrast to fixed-length subnetting, which allocates mask lengths based on the number of hosts in each campus /16, thus wasting address space in each campus /16.

3.2.1 VLSM Design Process

VLAN ID	VLAN Name	Subnet	Mask	Gateway	Broadcast	Max Hosts
10	Students-A	10.1.10.0/22	255.255.252.0	10.1.10.1	10.1.13.255	1,022
20	Staff-A	10.1.20.0/23	255.255.254.0	10.1.20.1	10.1.21.255	510
30	Admin-A	10.1.30.0/24	255.255.255.0	10.1.30.1	10.1.30.255	254
40	Research-A	10.1.40.0/23	255.255.254.0	10.1.40.1	10.1.41.255	510
50	Guest-A	10.1.50.0/23	255.255.254.0	10.1.50.1	10.1.51.255	510
60	CCTV-A	10.1.60.0/24	255.255.255.0	10.1.60.1	10.1.60.255	254

80	Mgmt-A	10.1.80.0/24	255.255.255.0	10.1.80.1	10.1.80.255	254
90	Native-A	10.1.90.0/24	255.255.255.0	—	10.1.90.255	—
100	DMZ-A	10.1.100.0/24	255.255.255.0	10.1.100.1	10.1.100.255	254
110	Servers-Int	10.1.110.0/24	255.255.255.0	10.1.110.1	10.1.110.255	254
120	Finance-A	10.1.120.0/24	255.255.255.0	10.1.120.1	10.1.120.255	254

Table 13: Campus A VLAN and Subnet Allocation

In Step 1, you'll identify the host requirements and subnet masks.

The minimum number of hosts per VLAN is calculated based on current users plus 20% growth over three years. There are two more addresses reserved for network and broadcast addresses. The closest power of 2 to (hosts + 2) is found and the subnet mask is determined by subtracting the exponent from 32. For example, the Student VLAN requires 10,000 students + 20% growth = 12,000 hosts. Adding 2 gives 12,002, and the nearest power of 2 is 2^14 = 16,384 (/18). However, the size of the subnet is limited on the campus to keep the broadcast domain efficient, so a /22 (1,022 usable hosts per subnet) is assigned per building, and multiple /22 blocks are allocated within the campus /16 for horizontal expansion. The complete VLSM calculation is shown in Table 6:

VLAN	Campus Users	Hosts Needed (+20% + 2)	Nearest 2^n	Theoretical Mask	Practical Mask	Usable Hosts (Practical)
Students-A (VLAN 10)	10,000	12,002	16,384 (2^14)	/18	/22	1,022

Staff-A (VLAN 20)	1,200	1,442	2,048 (2^11)	/21	/23	510
Admin-A (VLAN 30)	250	302	512 (2^9)	/23	/24	254
Research-A (VLAN 40)	1,200	1,442	2,048 (2^11)	/21	/23	510
Guest-A (VLAN 50)	500	602	1,024 (2^10)	/22	/23	510
CCTV-A (VLAN 60)	50	62	64 (2^6)	/26	/24	254
Mgmt-A (VLAN 80)	20	24	32 (2^5)	/27	/24	254
DMZ-A (VLAN 100)	10	14	16 (2^4)	/28	/24	254
Servers-Int (VLAN 110)	15	20	32 (2^5)	/27	/24	254
Finance-A (VLAN 120)	5	8	16 (2^4)	/28	/24	254

Table 14: VLSM Host Requirement Calculation

Note: The Theoretical Mask column represents the pure VLSM calculation for campus-wide host counts, while the Practical Mask column reflects the per-building subnet size applied in the actual design. Practical allocations override theoretical calculations to maintain broadcast domain efficiency (~1,000 hosts maximum) and to create natural security boundaries at the /24 level for sensitive groups. Multiple /22 or /23 blocks are reserved per campus for horizontal building expansion. This follows the Tutorial 06

methodology of matching subnet size to operational constraints rather than purely mathematical host counts (Tian and Gao, 2024).

Campus B and Campus C share the same VLAN IDs and subnets, but with different prefixes. The satellite campus templates are given in Table 7.

The following table lists the VLANs assigned to each campus:

VLAN ID	VLAN Name	Campus B Subnet	Campus C Subnet	Purpose
10	Students-B/C	10.2.10.0/22	10.3.10.0/22	Student data and devices
20	Staff-B/C	10.2.20.0/23	10.3.20.0/23	Academic and administrative staff
40	Research-B/C	10.2.40.0/23	10.3.40.0/23	Research laboratories and HPC access
50	Guest-B/C	10.2.50.0/23	10.3.50.0/23	Visitor and recreational WiFi
80	Mgmt-B/C	10.2.80.0/24	10.3.80.0/24	Switch and router management
90	Native-B/C	10.2.90.0/24	10.3.90.0/24	Native trunk VLAN (unused data)

Table 15: Campus B and C VLAN Allocation Templates

Core, distribution and WAN devices are connected via point-to-point links with /30 subnets to save address space. Each campus has a dedicated 10.x.250.0/24 block that

is used for this purpose, with 64 usable /30 subnets per campus. The Campus A link allocation is shown in Table 8:

Link Purpose	Subnet	Device A IP	Device B IP
Core-A1 ↔ Dist-A-Acad (Primary)	10.1.250.0/30	10.1.250.1	10.1.250.2
Core-A2 ↔ Dist-A-Acad (Backup)	10.1.250.16/30	10.1.250.17	10.1.250.18
Core-A1 ↔ Dist-A-Admin (Primary)	10.1.250.4/30	10.1.250.5	10.1.250.6
Core-A2 ↔ Dist-A-Admin (Backup)	10.1.250.20/30	10.1.250.21	10.1.250.22
Core-A1 ↔ Dist-A-DC (Primary)	10.1.250.8/30	10.1.250.9	10.1.250.10
Core-A2 ↔ Dist-A-DC (Backup)	10.1.250.24/30	10.1.250.25	10.1.250.26
Core-A1 ↔ Dist-A-Sec (Primary)	10.1.250.12/30	10.1.250.13	10.1.250.14
Core-A2 ↔ Dist-A-Sec (Backup)	10.1.250.28/30	10.1.250.29	10.1.250.30
Core-A1 ↔ R-A-WAN	10.1.250.32/30	10.1.250.33	10.1.250.34
Core-A2 ↔ R-A-WAN	10.1.250.36/30	10.1.250.37	10.1.250.38
Core-A1 ↔ FW-A-Edge	10.1.250.40/30	10.1.250.41	10.1.250.42
Core-A2 ↔ FW-A-Edge	10.1.250.44/30	10.1.250.45	10.1.250.46

Table 16: Campus A Point-to-Point Link Allocation (10.1.250.0/24)

3.2.1 Packet Tracer Validation

Cisco Packet Tracer was used to validate the above addressing scheme. A representative topology was designed and built, consisting of Campus A (Core-A1, Dist-A-Acad, Acc-A-Acad-1), Campus B (Core-B1, Acc-B-Teach-1), Campus C (Core-C1, Acc-C-Teach-1), and an inter-campus WAN router (R-A-WAN). All VLANs, SVI interfaces and static IP assignments were configured as per Tables 6-8.

The complete Packet Tracer topology with all devices connected and port statuses green is shown in Figure 1 (below), which represents the physical topology of the representative network.

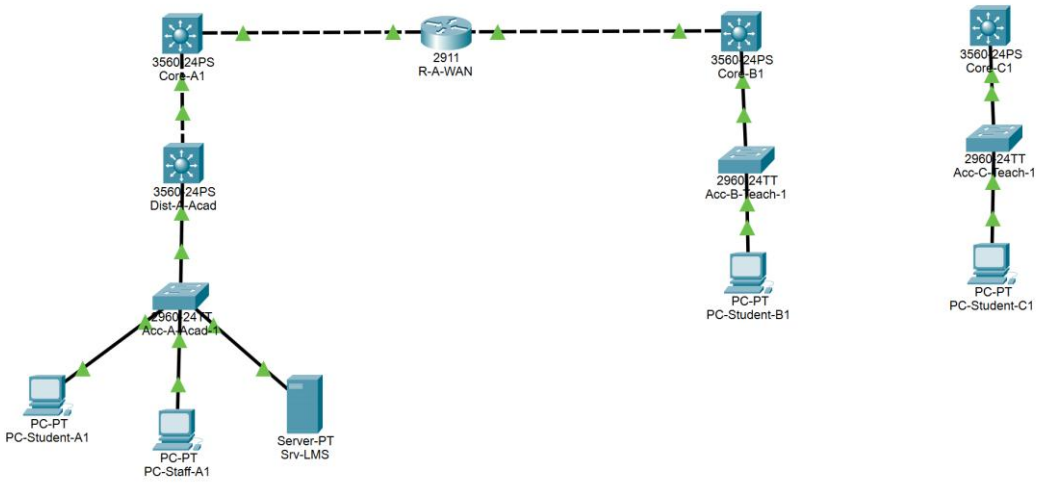


Figure 2: Representative Packet Tracer Topology: Multi-Campus Validation Model

Figure 2 shows that the SVI interfaces on the distribution switch (Dist-A-Acad) are administratively up with correct IP addresses for VLANs 10, 20, and 110. The output of the command "show ip interface brief" confirms that all the gateway addresses configured in Table 5 are active and reachable.

```
Dist-A-Acad#show ip interface brief | include Vlan
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 10.1.10.1       YES manual up                    up
Vlan20                 10.1.20.1       YES manual up                    up
Vlan110                10.1.110.1      YES manual up                    up
Dist-A-Acad#
```

Figure 3: SVI Interface Status- Distribution Switch (Dist-A-Acad)

The VLAN database on the access switch (Acc-A-Acad-1) shows that VLANs 10, 20, 110, and 90 exist and have been named as per the allocation plan:

```
Acc-A-Acad-1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Gig0/1, Gig0/2
10   Students-A                       active    Fa0/1
20   Staff-A                          active    Fa0/2
90   Native-A                         active
110  Servers-Int                      active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Figure 4: VLAN Database- Access Switch (Acc-A-Acad-1)

End-to-end validation is given in figure 4. A PC in VLAN 10 (PC-Student-A1, 10.1.10.10) successfully pings its own gateway (10.1.10.1), the Staff gateway in VLAN 20 (10.1.20.1), and the LMS server in VLAN 110 (10.1.110.10). The "Reply from" responses for TTL=255 (local hop) and TTL=127 (cross-VLAN via routing) indicate that the addressing, VLAN segmentation, and inter-VLAN routing are working correctly.

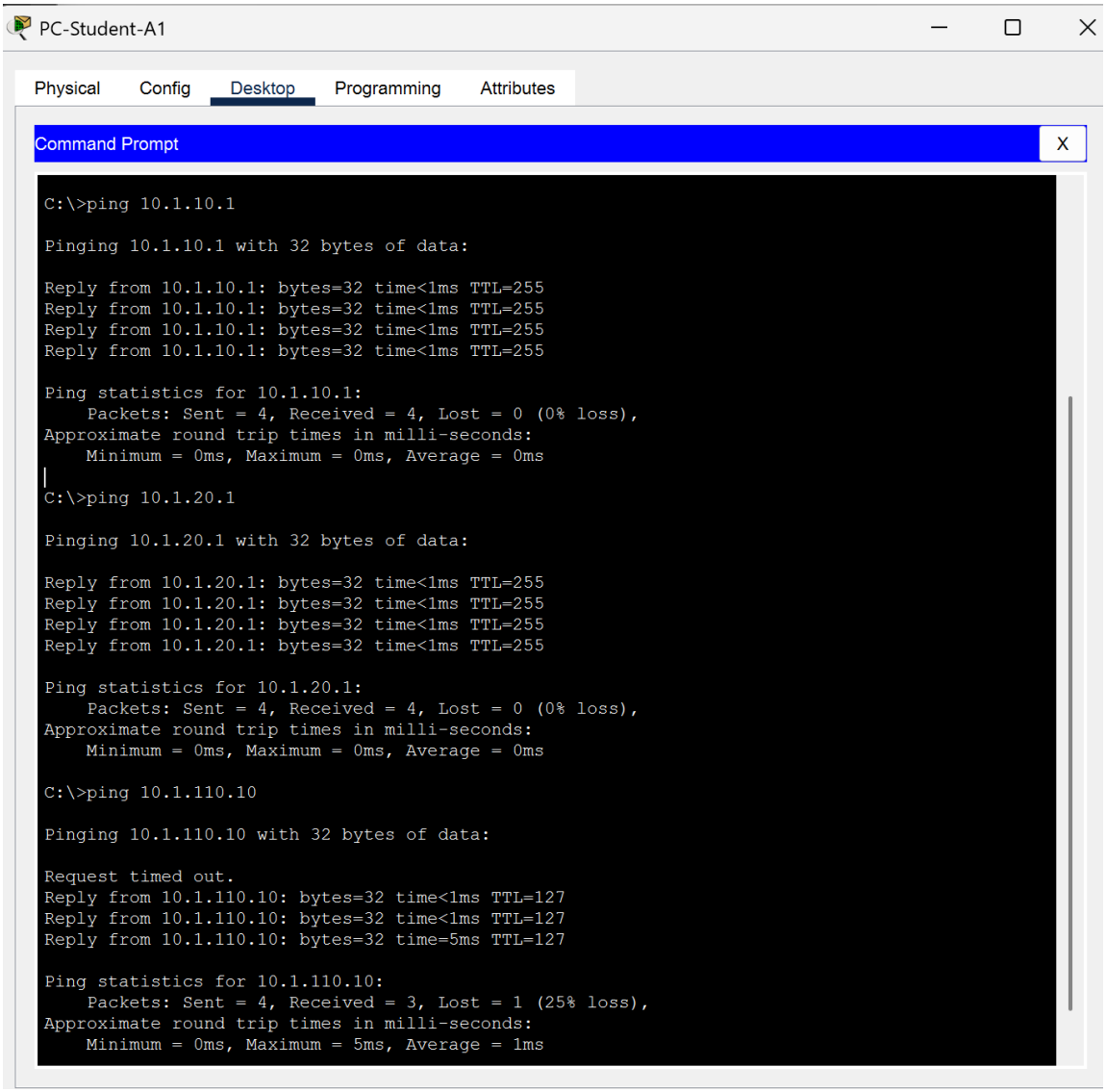


Figure 5: End-to-End Ping Validation- Cross-VLAN Routing Test

3.3 Addressing Hierarchy and Scalability.

The addressing scheme is hierarchical and follows the physical campus topology and administrative boundaries, with four levels:

Level 1– Campus Boundary (/16): The first two octets (10.x) define the campus. This allows route summarisation at the WAN edge, with just one OSPF area or BGP advertisement per campus, which simplifies the routing table. For example, all Campus A routes are summarised as 10.1.0.0/16.

Level 2– Functional Zone (/22 to /24): The third octet is used to identify devices by function (students, staff, servers, management). This logical grouping makes it easier to create access control lists (ACLs), define firewall rules, and apply QoS policies. For example, the traffic from 10.1.10.0/22 (Students) can be rate limited or prioritised as a whole.

Level 3– Building/Floor (Host ID). The fourth octet is used for individual devices. Each Student /22 subnet is divided into four /24 blocks, representing different buildings or floors. For example, in the first /22 (10.1.10.0/22), addresses 10.1.10.1–10.1.10.254 serve Building 1, 10.1.11.1–10.1.11.254 serve Building 2, 10.1.12.1–10.1.12.254 serve Building 3, and 10.1.13.1–10.1.13.254 serve Building 4. Additional /22 blocks (e.g., 10.1.14.0/22, 10.1.18.0/22) are reserved for further academic buildings. This architecture allows for location-specific troubleshooting and address predictability for IT support personnel.

Level 4– Point-to-Point Links (/30): Inter-device links are dedicated /30 subnets to reduce waste. Each /30 will use only 2 usable addresses, and there will be 62 more /30 subnets in the 10.x.250.0/24 block to expand on.

Address space is also intentionally reserved, providing further scalability. Campuses D and E are assigned to blocks 10.4.0.0/16 and 10.5.0.0/16 respectively, with the remaining blocks 10.6.0.0/16 to 10.15.0.0/16 being available for future campus or centralised services. VLAN 90 (Native) and management VLAN 80 are standardised within each campus, providing consistent trunking and out-of-band management practices throughout each campus.

IPv6 Stateless Address Autoconfiguration (SLAAC) is used for end-user devices, which decreases the load on DHCP servers and makes it easier for mobile devices to connect (Tutorial 06, Section 6.5). SLAAC works in three phases: (1) The host generates a link-local address (fe80::/64) and checks for uniqueness using Neighbor Solicitation; (2) The host sends a Router Solicitation to find out the network prefix; (3) The host uses the network prefix and its interface identifier (EUI-64 or privacy extension) to create a global address. SLAAC is "stateless" because there is no central server that keeps track of address assignments, which is a benefit to the university's 20,000+ devices. Stateful DHCPv6 is used for servers and IoT devices that need a fixed address.

The scalability of IPv6 is the same. Each campus /48 is subdivided into /64 VLAN subnets, with fd00:x:10::/64 corresponding to VLAN 10, fd00:x:20::/64 to VLAN 20, and so on. The large address space removes the need for NAT in the campus, making peer-to-peer applications and IoT deployment easier. The router loopbacks and inter-campus links are assigned to fd00:ff::/48, which provides a separate management and transit plane from user traffic.

3.4 Addressing Design Justification

The VLSM allocation in Table 5 and 6 is based on the systematic approach outlined in Tutorial 06 (Tian and Gao, 2024): (1) calculate host requirements such as growth margin and network/broadcast addresses; (2) sort the subnets in order of size (largest first) to avoid fragmentation; (3) allocate addresses sequentially from the campus block; (4) check that the total usage does not exceed the /16 boundary. This methodology is applied to each subnet size in the following subsections to justify the size.

3.4.1 Student VLAN (/22)

Campus A is home to about 10,000 students in several academic buildings. Following Tutorial 06 Step 1, the requirement is 10,000 + 20% growth + 2 = 12,002 hosts, requiring 2^14 = 16,384 addresses (/18). For performance, however, the size of the broadcast domain is limited to ~1,000 hosts (Tian and Gao, 2024). Therefore, a /22 subnet (1,022 usable hosts) is allocated per building, with multiple /22 blocks (e.g., 10.1.10.0/22, 10.1.14.0/22, 10.1.18.0/22) reserved for current and future academic buildings. With 10,000 students and expected growth of 20%, the Campus A /16 supports up to 64 /22 subnets, giving the campus plenty of room to grow. This is a compromise between address availability and broadcast control, and adheres to the Tutorial 06 rule of matching subnet size to operational constraints.

3.4.2 Staff and Research VLANs (/23)

Academic staff (1200 at Campus A) and research labs need 510 hosts each with a /23. This size is a compromise between address availability and broadcast control. Research VLANs contain high bandwidth workstations and lab equipment, and the size of the

subnet is kept to a moderate size to ensure that excessive broadcast traffic does not affect latency sensitive instruments.

3.4.3 Administrative and CCTV VLANs (/24)

The administrative staff (250 users) and CCTV/security systems (50 devices) each need less than 254 addresses per VLAN. Each function (VLAN 30 and VLAN 60) has its own VLAN, giving enough capacity with 20% growth margin, and preserving security isolation between administrative and surveillance traffic. For these smaller groups, with security concerns, a /24 is suitable. The /24 provides a natural boundary for firewall policies: For instance, the Finance VLAN (120) is isolated at the /24 level and has strict ACLs that block any inbound traffic from the Student or Guest VLANs.

3.4.4 DMZ and Internal Server VLANs (/24)

Each of the DMZ (VLAN 100) and Internal Servers (VLAN 110) has /24 subnets. Public facing services (university website, LMS gateway, email relay) are located on the DMZ and a separate subnet is needed for the firewall DMZ interface configuration. Internal Servers (LMS, DNS, DHCP, student records) are on a separate /24 network with gateway 10.1.110.1, allowing the East-West firewall to inspect traffic between the DMZ and internal networks.

3.4.5 Point-to-Point Links (/30)

There are exactly 2 addresses used per inter-device link (one for each endpoint). Following Tutorial 06, hosts + 2 = 4, so 2^2 = 4 addresses are needed, giving a /30 mask (32 − 2 = 30). If legacy equipment and routing protocols do not support /31 subnets, then use /30. The 10.x.250.0/24 block gives 64 /30 subnets per campus, which includes current links as well as a lot of future expansion (additional distribution switches, WAN routers, or firewall pairs) without re-addressing, as described in Tutorial 06 Step 4 (verification and reservation).

3.4.6 IPv6 Dual-Stack Justification

The dual-stack approach enables the university to keep running IPv4 services and slowly migrate to IPv6. To guarantee the stability of the addresses in case the university switches ISPs, internal use of ULA blocks (fd00::/48) is avoided. The one-to-one relationship

between IPv4 VLANs and IPv6 /64s makes it easier for IT personnel to learn and helps minimize configuration mistakes. If the university is ready to receive public IPv6 services, then the university can deploy Global Unicast prefixes in parallel with the existing ULA without affecting internal routing.

3.4.7 Reserved Space and Future Expansion

The design reserves 10.4.0.0/16 and 10.5.0.0/16 for future campuses, and 10.6.0.0–10.15.0.0/16 for additional expansion. Within each campus, the 10.x.128.0/17 block is reserved for future building additions. This reservation allows for new campuses or buildings to be added without overlapping existing subnets or having to re-address deployed devices. This expansion is facilitated by the routing protocol selection (OSPF) which includes area addition and automatic route propagation, as described in Section 4.

4. Network Routing Architecture

4.1 Routing Protocol Selection

The Greenfield University multi-campus network requires robust and scalable routing architecture to support the hierarchical core-distribution-access design and meet the performance objectives established in Stage 1. Two routing protocols are selected for the design: OSPF (Open Shortest Path First) as the Interior Gateway Protocol (IGP) for internal routing, and BGP (Border Gateway Protocol) for external internet connectivity.

4.1.1 OSPF Selection & Justification

OSPF is chosen as the internal routing protocol for the following reasons:

Reason	Explanation
Scalability and Hierarchical Support	OSPF is designed with multi-area support, which is essential for the university's three-campus architecture. The protocol's ability to segment the network into areas reduces the size of the topology database that each router must maintain, ensuring manageable memory and CPU utilization. The design implements Campus A as Area 0 (backbone), with Campus B and Campus C configured as separate non-backbone areas, allowing for future campus additions without redesign
VLSM and Route Summarization Support	OSPF supports Variable Length Subnet Masking (VLSM), which is critical for the hierarchical addressing scheme designed in Section 3.2. The protocol enables route summarization at Area Border Routers (ABRs), reducing routing table sizes and improving convergence times. For example, all routes from Campus B can be summarized in single 10.2.0.0/16 advertisement.
Fast Convergence	OSPF provides sub-second convergence through its link-state mechanism and support for Bidirectional

	Forwarding Detection (BFD). This directly supports the Stage 1 requirement of <5 second failover for redundant links
Open-Standard and Vendor Neutral	Unlike EIGRP, which is a Cisco proprietary protocol, OSPF is an open standard defined in RFC 2328 [2] [3]. This aligns with the design assumption to "avoid proprietary features where open-standard alternatives exist" (assumption R8) and ensures flexibility in vendor selection.

Table 17: OSPF Selection & Justification

Why OSPF over EIGRP:

While research indicates that EIGRP may offer faster convergence times (9 seconds) compared to OSPF (30 seconds) in some configurations, OSPF demonstrates better packet delay performance [3]. More importantly, OSPF's open-standard nature eliminates vendor lock-in concerns, which is a critical requirement for a university seeking long-term flexibility and competitive procurement [3]. OSPF is also significantly more scalable than EIGRP for large enterprise deployments [3].

Why OSPF over RIP:

RIP's hop-count limit of 15, slow convergence, and lack of support for VLSM make it unsuitable for a network of this scale and complexity [3].

4.1.2 BGP Selection & Justification

BGP is selected for the internet edge to manage connectivity to the dual ISP connections (ISP 1: 5 Gbps primary, ISP 2: failover) [4].

- **Purpose**: BGP is the standard Exterior Gateway Protocol (EGP) designed for routing between autonomous systems. While OSPF is designed to carry a few thousand routes within a single domain, BGP is designed to carry significantly more routes [4], making it appropriate for managing full internet routing tables [5].

- **Redundancy & Failover**: BGP enables automatic failover between the two ISP links by detecting the loss of BGP neighbors and withdrawing/advertising routes accordingly [5]. This supports the Stage 1 requirement for dual ISP redundancy.
- **Policy Control**: BGP's path-vector architecture allows the university to implement routing policies based on business preferences, such as preferring the primary ISP link unless it fails, and controlling inbound and outbound traffic flows [5].

4.2 Routing Design & Policies

4.2.1 OSPF Configuration & Design

OSPF implementation follows these design principles:

Design Principle	Explanation
Multi-Area Design	Area 0 serves as the backbone at Campus A core. Campus B and C are configured as separate OSPF areas to localize routing updates. This approach reduces LSA flooding and improves scalability [2] [5].
Route Summarization	ABRs at Campus A will summarize routes before advertising them to other areas. For example, Campus B's 10.2.0.0/16 block will be advertised as a single summary route, minimizing routing table entries across the network.
Stub Areas	Campus B and C will be configured as Totally Stubby Areas to further reduce routing table sizes. These areas will receive a default route from Campus A rather than external or inter-area routes, significantly reducing memory and processing requirements on distribution switches.
Router IDs	Loopback interface IP addresses will be used as Router IDs for stability and reliability [4].

Table 18: OSPF Configuration & Design

4.2.2 BGP Configuration & Policies

- **Autonomous System Number**: The university will obtain a public ASN for peering with both ISPs.
- **Prefix Advertisement**: The university will advertise its public IP prefixes to both ISPs, with appropriate AS-path prepending on the backup link to make it less preferred for inbound traffic.
- **Local Preference**: LOCAL_PREF will be used to set the primary ISP as the preferred path for outbound traffic [5].
- **Route Redistribution Management**: To avoid the risks associated with redistributing BGP routes into OSPF [4], a default route (0.0.0.0/0) will be injected from BGP into OSPF. Internal routers will use this default route for internet-bound traffic, while OSPF will carry only internal network prefixes [4]. Redistribution filtering using ACLs and OSPF max-lsa protection will be implemented to prevent route leaks [4].

4.2.3 OSPF-BGP Interaction

- **Redistribution Boundary**: Redistribution between OSPF and BGP will occur only at the Autonomous System Boundary Routers (ASBRs) connected to the internet edge firewalls.
- **Protection Mechanisms**: OSPF max-lsa configuration will be implemented on all OSPF devices to protect against excessive LSA reception, with thresholds set based on the campus route counts [4].

4.3 Redundancy & Failover Strategy

- **OSPF ECMP**: Equal Cost Multi-Path (ECMP) will be enabled on OSPF to load-balance traffic across the redundant 10 Gbps inter-campus links. This ensures maximum utilization of available bandwidth and rapid failover [4].
- **BGP Failover:** BGP will detect ISP link failures and automatically switch to the backup connection. The convergence time is expected to be within 3-5 minutes, which is

acceptable for internet connectivity while satisfying the <5 second requirement for internal campus routing [5].

- **BFD Integration:** Bidirectional Forwarding Detection will be deployed with OSPF for sub-second link failure detection, enabling rapid convergence of internal routing [4].

4.4 Routing Architecture Justification

The selected OSPF and BGP architecture directly addresses the critical challenges identified in Stage 1:

- Challenge C6 (Static routing only): The dynamic OSPF and BGP implementation eliminates manual static route management and provides automatic path selection and failover.
- Challenge C4 (Single points of failure): ECMP and BGP failover provide automatic redundancy across the redundant network links and dual ISP connections.
- Challenge C1 (Insufficient bandwidth): ECMP enables load-sharing across the redundant 10 Gbps links, increasing effective inter-campus bandwidth and meeting the 10-12 Gbps aggregate demand.
- Scalability Goals: the multi-area OSPF design supports the addition of up to two new campuses as new areas without network-wide reconfiguration [2] [3]. Reserved address blocks and route summarization ensure the network can grow without major redesign.

5. Network Performance Architecture

5.1 Scalability & Performance Strategy

5.1.1 Scalability Design

Network architecture ensures scalability through the following mechanisms:

- **Hierarchical Three -Tier Model**: The core-distribution-access architecture, as detailed in Section 2, provides inherent scalability [2] [6]. Each layer can be expanded independently: access switches can be added to support new buildings, distribution switches can be added to support new functional blocks, and core switches can be upgraded for higher capacity without redesigning the entire network [6].
- **Modular Design**: The modular approach enables “pay-as-you-grow” expansion [6]. Each building at Greenfield University will be served by a separate distribution layer, providing flexibility in adding new buildings with minimal impact on existing ones and facilitating fault isolation [6].
- **Scalable Addressing**: The IPv4 /16 per campus with VLSM allocation and reserved IPv4 and IPv6 blocks ensures addressing capacity for 20% annual user growth and two additional campuses without re-addressing [2].
- **Routing Scalability**: OSPF multi-area design localizes routing updates, while route summarization reduces routing table sizes, enabling the network to scale to support additional campuses as new OSPF areas [2].

5.1.2 Performance Strategy

- **Bandwidth Provisioning**: The design provides 10 Gbps inter-campus links (addressing Challenge C1) and 1 Gbps access layer connections, meeting the peak aggregate demand of 10-12 Gbps [2].
- **Latency Optimization**: OSPF selects the shortest path, keeping the end-to-end latency between campuses and the central data centre below 50 ms as specified in Section 1.2.5.

- **Throughput**: The hierarchical design with ECMP enables full utilization of redundant links, ensuring high throughput for bandwidth-intensive applications such as research data transfers and video conferencing.

5.2 QoS Architecture & Traffic Prioritization

5.2.1 Differentiated Services Model

The university will implement a Differentiated Services (DiffServ) architecture as defined in RFC 2475 [7] [8] [9]. DiffServ is chosen over the Integrated Services (IntServ) model because it is more scalable and efficient for large enterprise networks [9]. While IntServ provides absolute guarantees via resource reservation (RSVP), its operational complexity is unsuitable for a network of this scale [9]. DiffServ, by contrast, classifies traffic into classes and provides different treatment to each class, offering a more flexible and manageable approach [9].

5.2.2 Classification & Marking

Classification and DSCP (Differentiated Services Codepoint) marking will be performed at the access layer switches as traffic enters the network [7] [8]:

- Trust Boundary: The trust boundary is established at the access switch. All DSCP markings from end-user devices will be overwritten to prevent abuse.
- Classification Methods: Access Control Lists (ACLs) and Network-Based Application Recognition (NBAR) will identify application traffic based on IP addresses, protocols, and port numbers [7].

Traffic Class	DSCP Marking	Queue Type	Bandwidth Allocation	Use Case
Network Control	CS6 (48)	Strict Priority	5%	OSPF, BGP, STP, routing protocols
Real-Time (Voice/Video)	EF (46)	Low Latency Queue (LLQ)	30%	Video conferencing,

				VoIP, live streaming
Critical Data	AF41 (34)	Weighted Fair Queue (WFQ)	35%	LMS interface, financial database, student records
Transactional Data	AF21 (18)	Weighted Fair Queue	15%	Email, library catalogue, web browsing
Best Effort	BE (0)	Default FIFO	10%	Guest internet, recreational traffic
Scavenger	CS1 (8)	Default FIFO	5%	Backups, P2P file sharing, software updates

Table 19: Traffic Classes & DSCP Markings

5.2.3 Queuing & Scheduling

- **Low Latency Queue (LLQ):** EF-class traffic (Real-Time) will be transmitted through a strict priority queue. The LLQ is rate-limited to 30% of link capacity to prevent starvation of other traffic classes. This protects real-time teaching and collaboration tools from being degraded by other traffic.
- **Weighted Fair Queuing (WFQ):** Assured Forwarding (AF) queues will use WFQ with the bandwidth percentages specified in the table above, ensuring minimum bandwidth guarantees for each class.
- **Congestion Avoidance – Weighted Random Early Detection (WRED):** WRED will be enabled on all AF queues to proactively completely full. This prevents global TCP synchronization and maintains throughput for higher-priority flows within the same class.

5.2.4 Policing & Shaping

- **Ingress Policing:** Guest VLAN traffic will be policed to a maximum of 2 Gbps at the internet edge to prevent recreational traffic from consuming all internet bandwidth.
- **Egress Shaping:** Traffic shaping will be configured on inter-campus links to smooth bursty research traffic, reducing the likelihood of tail drops.

5.3 Service Level Agreements (SLAs)

Service Level Agreements define measurable performance commitments between the IT department and university stakeholders:

Service/Application	Metric	Target (SLA)	Detection/Mitigation
LMS Access	End-to-End Latency	< 50ms	NetFlow monitoring; QoS policy adjustment if exceeded
Video Conferencing	Jitter	< 30ms	IP SLA monitoring; EF queue bandwidth increases if exceeded
High-Priority Traffic	Packet Loss (peak hours)	< 1%	WRED and traffic policing exceeded
Mission-Critical Systems	Availability	99.99% (< 52 mins/year downtime)	Redundant hardware and routing failover
Inter-Campus Traffic	Throughput	10 Gbps (per link)	SNMP link utilization monitoring; capacity planning
Video Conferencing	Concurrent Sessions	Support 5,000 sessions	QoS monitoring; capacity upgrades planned at 80% utilization

Table 20: SLAs

5.3.1 SLA Monitoring & Verification

- **SNMP:** Monitors interface utilization, CPU, memory, and queue drops.
- **NetFlow**: Analysis traffic patterns and identifies top talkers.
- **IP SLA**: Measures end-to-end latency and jitter for EF traffic to verify QoS effectiveness.
- **Regular Audits:** Monthly QoS audits to ensure DSCP markings are preserved end-to-end and queuing policies are operating as configured.

5.4 Performance & Availability Justification

The performance architecture directly addresses the challenges and objectives identified in Stage 1:

- **Challenge C3 (Absence of QoS):** The detailed DiffServ model with DSCP marking, LLQ, and WRED provides comprehensive traffic prioritization, ensuring mission-critical applications (LMS, video conferencing, financial systems) receive priority over recreational traffic during congestion [9].
- **Challenge C1 (Insufficient bandwidth):** The 10 Gbps inter-campus backbone with ECMP load balancing provides the capacity required for the 10-12 Gbps peak aggregate demand.
- **Goal 5 (Support 5,000 concurrent video sessions):** The QoS architecture, with its 30% LLQ allocation for EF traffic and underlying 10 Gbps capacity, provides the bandwidth and prioritization necessary to support 5,000 concurrent video conference sessions with acceptable quality (<100 ms latency, <30 ms jitter).
- **Scalability Goals:** The hierarchical design, OSPF multi-area routing, and reserved addressing ensure the network can grow to support 20% annual user growth and two additional campuses without major redesign [2] [4] [6].
- **Performance and Reliability:** The combination of redundant hardware, OSPF/BGP failover mechanisms. QoS traffic prioritization, and formal SLAs transform the network form a source of frustration into a reliable foundation for teaching, research, and administration.

6. Network Management Architecture

The proposed multi-campus university network requires a comprehensive network management architecture to ensure reliable operation, rapid fault detection, and efficient troubleshooting. Due to the large number of users, applications, and network devices distributed across multiple campuses, proactive monitoring and centralized management are essential for maintaining network availability, performance, and security. The network management architecture incorporates monitoring and visibility tools, fault management procedures, and centralized logging and event management to support day-to-day operations.

6.1 Monitoring and Visibility Framework

The proposed multi-camps university network will implement a centralized network management architecture to provide continuous visibility into network performance, availability, and resource utilization across all campuses. Given the large number of users, applications and interconnected sites, proactive monitoring is essential to maintain service quality and ensure that the performance objectives identifed in Stage 1 are achieved.

6.1.1 SNMP-Based Device Monitoring

Simple Network Management Protocol (SNMP) will be deployed on all core routers, distribution switches, firewalls, wireless controllers, and critical servers throughout the university network. SNMP provides a standardized method for collecting operational data from network devices and enables centralized monitoring through the Network Management System (NMS)

The protocol will be used to monitor device health and performance metrics, including CPU utilization, memory consumption, interface utilization, link status, and queue drops. By continuing to collect this information, network administrators can identify abnormal behaviour, capacity limitations, and potential hardware failures before they affect network services.

Threshold-based monitoring will also be implemented to generate alerts when predefined performance limits are exceeded. This proactive approach helps maintain

service availability and supports efficient resource planning as the university network expands.

6.1.2 NetFlow Traffic Visibility

NetFlow will be enabled on core routing devices to provide detailed visibility into network traffic patterns and application usage. Unlike SNMP, which focuses on device performance, NetFlow provides information about the actual traffic traversing the network.

NetFlow data will be used to identify high-bandwidth applications, monitor user traffic patterns, detect abnormal network behaviour, and support capacity planning activities. This capability is particularly important for managing LMS traffic, video conferencing sessions, cloud-based services, research data transfers, and inter-campus communications.

Traffic analysis provided by NetFlow will assist administrators in understanding bandwidth consumption trends and making informed decisions regarding future network upgrades and resource allocation

6.1.3 IP SLA Performance Monitoring

In addition, IP SLA monitoring will be implemented to measure end-to-end network performance for critical university applications and services. IP SLA generates synthetic traffic to actively test network conditions and verify application performance.

Key performance indicators measured by IP SLA include latency, jitter, packet loss, and response times. These metrics are particularly important for delay-sensitive applications such as video conferencing, Voice over IP, and cloud-based collaboration platforms.

The collected performance data will be compared against the Service Level Agreements (SLAs) and QoS requirements defined in the network performance architecture. This ensures that network services consistently meet the expected performance standards for students, academic staff, and administrative users.

6.1.4 Centralized Network Management System

A centralized Network Management System (NMS) will be deployed to aggregate monitoring information from all campuses into a single management platform. The NMS serves as the primary interface for the network administrators and provides real-time visibility into network operations.

The platform will collect information from SNMP, NetFlow, Syslog, and IP SLA monitoring systems to generate dashboards, performance reports, and automated alerts. Administrators will be able to monitor network status, identify developing issues, and analyze historical trends through a unified management interface.

Centralized management simplifies operational processes, reduces administrative overhead, and enables consistent monitoring policies across all campuses.

Function	Technology	Purpose
Device Monitoring	SNMP	Monitor CPU, memory, interfaces, and device health
Traffic Analysis	NetFlow	Analyse traffic patterns and bandwidth usage
Performance Monitoring	IP SLA	Measure latency, jitter, and packet loss
Logging	Syslog	Centralized log collection
Alerting	NMS Platform	Automated notifications and fault detection
Auditing	QoS Auditing	Verify SLA compliance and QoS effectiveness

Table 21. Network Management Tools

6.2 Fault, Logging, and Event Management

To ensure high network availability and reliability, the proposed architecture incorporates centralized fault, logging, and event management processes. These

mechanisms enable administrators to rapidly identify, diagnose, and resolve network issues while maintaining service continuity across all campuses.

6.2.1 Fault Management Strategy

Fault management will rely on automated monitoring and threshold-based alerts generated through the NMS. Device failures, interface outages, excessive bandwidth utilization, routing instability, and abnormal performance conditions will automatically trigger notifications to network administrators.

The architecture supports proactive fault detection by continuously monitoring network devices and services. When a fault is detected, administrators can quickly identify the affected components and behind corrective actions before significant service disruption occurs.

Network resilience is further improved through redundant hardware, OSPF failover mechanisms, and dual WAN connectivity between campuses. These features help minimize downtime and maintain network availability during equipment failures or link outages.

Stage	Activity	Outcome
Detection	SNMP/IP SLA identifies normal condition	Alert generated
Notification	NMS sends alert to administrators	Incident created

Diagnosis	Logs and monitoring data reviewed	Root cause identified
Resolution	Corrective action implemented	Service restored
Verification	Performance re-tested	Issues confirmed resolved
Documentation	Incident recorded	Future reference and auditing

Table 22. Fault Management Workflow

6.2.2 Centralized Logging Architecture

Centralized logging will be implemented during a Syslog server that collects logs from routers, switches, firewalls, servers, and security appliances across all campuses.

The Syslog infrastructure provides a consolidated repository of operational and security-related events, allowing administrators to analyse historical activity investigate incidents more efficiently. Centralized logs also support compliance requirements, auditing processes, and long-term operational analysis.

By storing logs in a central location, the university can improve troubleshooting efficiency and maintain a complete record of network activity for future reference.

6.2.3 Event Correlation and Analysis

Event management will correlate SNMP alerts, Netflow information, IP SLA measurements, and Syslog events to provide a comprehensive view of network operations.

Rather than examining individual alerts separately, event correlation allows administrators to identify relationships between multiple events and determine the root cause of network issues more quickly. This reduces troubleshooting time and improves incident response effectiveness.

Event correlation also assists in identifying abnormal traffic patterns, potential security incidents, and recurring performance issues that may require long-term remediation.

6.2.4 Network Auditing and Continuous Improvement

To maintain long-term service quality, monthly network performance and QoS audits will be conducted to verify SLA compliance and evaluate the effectiveness of existing network policies.

These audits will review bandwidth utilization, traffic prioritization effectiveness, device performance metrics, and service availability records. Historical monitoring data will be analysed to identify capacity trends and potential bottlenecks before they affect users.

The auditing process supports continuous improvements by providing recommendations for optimization, infrastructure upgrades, and future expansion planning. This ensures that the network remains capable of supporting increasing user demands and new applications over time.

6.3 Network Management Design Justification

The proposed network management architecture supports the scalability, performance, and reliability requirements identified during Stage 1. SNMP provides efficient monitoring of network devices and infrastructure health, while NetFlow delivers detailed visibility into traffic patterns and bandwidth consumption. IP SLA monitoring further enhances operational visibility by validating latency, jitter, and packet loss requirements for critical services.

Centralized fault management, logging, and event correlation reduce troubleshooting time and improve operational efficiency across all campuses. Automated alerting enables proactive issue resolution before users experience service degradation.

The architecture also complements the QoS and SLA framework presented int he Network Performance Architecture by continuously monitoring key performance indicators and verifying compliance with established service targets. Together, these capabilities provide a scalable and manageable network environment capable of supporting future growth, additional campuses, and increasing application demands without significant redesign.

6.3.1 Alignment with State 1 Requirements

The proposed network management architecture directly supports the requirements identified during Stage 1 of the project. The large user population, critical academic applications, research services, and administrative systems require continuous monitoring to ensure reliable operation and acceptable performance levels.

The monitoring framework supports the network's bandwidth, availability, QoS, and scalability requirements by providing visibility into resource utilization and service performance. In addition, centralized management allows administrators to maintain consistent operational policies across all campuses while supporting future growth and expansion.

By continuously monitoring network conditions and application performance, the university can ensure that critical services such as the LMS, video conferencing platforms, cloud services, and financial systems meet their defined service requirements.

6.3.2 Benefits of the Proposed Architecture

The proposed network management architecture provides several operational and strategic benefits for the university. Continuous monitoring enables proactive identification of issues before they impact users, reducing downtime and improving service reliability.

Centralized management simplifies administration by consolidating monitoring, logging, reporting, and alerting functions into a unified platform. This improves troubleshooting efficiency and reduces the time required to diagnose and resolve incidents.

The architecture also supports long-term scalability by providing the visibility and operational control necessary to accommodate additional campuses, users, applications, and network devices. As a result, the university can maintain a secure, high-performance, and manageable network environment while supporting future technological growth and evolving educational requirements.

7. Security Architecture

The proposed multi-campus university network requires comprehensive, layered security architecture designed to protect academic, administrative, research, and cloud-based services from internal and external cyber threats. Due to the distributed nature of the network and the large number of users across multiple campuses, the security design adopts a defense-in-depth model combined with Zero Trust principles, ensuring that no user, device, or network segment is inherently trusted.

This architecture focuses on maintaining confidentiality, integrity, and availability while supporting secure access to shared resources, remote learning environments, and mission-critical systems such as LMS platforms, financial databases, and research applications.

7.1 Network Segmentation Strategy

Network segmentation is implemented as the foundational security layer to reduce attack surfaces, limit lateral movement, and improve traffic control across the university infrastructure.

7.1.1 VLAN-Based Segmentation Model

The network is logically divided into VLANs based on user roles, device types, and service functions. This ensures that traffic is isolated between groups with security requirements.

Each VLAN is assigned a specific security policy:

- Students are restricted to academic and internet services.
- Staff VLANs are granted controlled access to internal systems.
- Administrative and financial VLANs are placed under stricter control due to sensitive data.
- Guest VLANs is fully isolated from internal university systems.

This structure ensures that even if one segment is compromised, the attacker cannot easily access other parts of the network.

7.1.2 Critical Systems Isolation (Server Segmentation)

High-value systems such as:

- LMS
- Student information systems
- Payroll and finance systems
- Research databases

are placed in a dedicated server VLAN/secure data center zone.

This segment is heavily restricted using firewall rules and ACLs, allowing only authorized services and authenticated users to communicate with it. This reduces exposure to threats such as ransomware, insider attacks, and unauthorized data access.

7.1.3 DMZ Design for Public Services

A Demilitarized Zone (DMZ) is deployed between the internal network and the external internet.

Services hosted in the DMZ include:

- University website
- LMS web portal access
- VPN gateway
- Public research portals

The DMZ acts as a buffer zone. Even if a public-facing service is compromised, attackers cannot directly access internal systems due to firewall isolation.

7.1.4 Inter-VLAN Security Controls

Inter-VLAN routing is controlled using Layer 3 switches and firewall-based policy enforcement.

Traffic between VLANs is governed by:

- Role-based access control.
- Port-level restrictions.
- Application-level filtering (where applicable)

For example:

- Students -> LMS allowed
- Students -> Payroll denied
- Admin -> Finance allowed
- Guests -> Internet only

This enforces least privileged access, a core security principle in enterprise network design.

VLAN	User Group	Purpose
VLAN 10	Students	General student access
VLAN 20	Academic Staff	Teaching and research
VLAN 30	Administrative Staff	Business operations
VLAN 40	Management	Executive services
VLAN 50	Research Labs	Research applications
VLAN 60	Guest Users	Internet-only access
VLAN 70	Servers	Internal services
VLAN 80	Network Management	Monitoring systems

Table 23. VLAN Security Mapping

7.2 Firewall and Access Control Design

The firewall architecture is designed as a multi-layered defence system that protects both perimeter and internal segments of the university network.

7.2.1 Perimeter Firewall (Edge Security Layer)

Next-Generation Firewalls (NGFWs) are deployed at the network edge to inspect all incoming and outgoing traffic.

They provide:

- Deep packet inspection (DPI)
- Application-aware filtering (e.g., blocking risky apps)
- Intrusion prevention capabilities
- URL and content filtering

- Threat intelligence integration

This ensures that malicious traffic is blocked before entering the university network.

7.2.2 Internal Firewall Segmentation

Internal firewalls are placed between sensitive network zones such as:

- Admin <-> Student networks
- Finance <-> General staff networks
- Server farm <-> user VLANs

This ensures that even if an internal device is compromised, attackers cannot move freely across the network.

This design significantly reduces lateral movement risk, which is one of the biggest threats in modern enterprise networks.

7.2.3 Access Control Strategy (ACL+ RBAC)

Access Control Lists (ACLs) and Role-Based Access Control (RBAC) are used together:

ACLs enforce technical packet-level rules

RBAC enforces user identity-based permissions

Example policy rules:

- Allow HTTPS access to LMS from all student VLANs
- Deny database access from student VLANs
- Allow admin VLAN access to payroll servers
- Restrict management access to secure VPN only

7.2.4 Zero Trust Implementation

The architecture follows a Zero Trust model:

- No user is trusted by default
- Every access request is verified
- Continuous authentication is enforced

This includes:

- Multi-Factor Authentication
- Device validation
- Session monitoring

This is especially important in a university environment where users frequently connect from multiple devices and locations.

7.3 VPN, IDS/IPS, and Encryption Strategy

This section provides the advanced security layer responsible for secure communication, threat detection, and data protection.

7.3.1 Secure VPN Architecture

A Virtual Private Network (VPN) is implemented for:

- Academic staff remote teaching
- Administrative remote teaching
- Management access
- External research collaboration

VPN connections use:

- IPSec tunnels (site-to-site)
- SSL VPN (remote users)

All traffic is encrypted to prevent interception over public networks.

7.3.2 IDS/IPS Deployment Strategy

Intrusion Detection and Prevention Systems (IDS/IPS) are deployed at:

- Internet edge
- DMZ boundary
- Core network segments

IDS functions:

- Detect suspicious traffic patterns
- Generate alerts for administrators

IPS functions:

- Block malicious traffic in real time
- Prevent known attack signatures

This protects against:

- DDoS attacks
- Port scanning
- Malware propagation
- Unauthorized access attempts

7.3.3 Encryption Strategy

Encryption is applied at multiple layers:

- HTTPS for web applications.
- TLS for communication between services
- SSH for device management
- IPSec for VPN tunnels

This ensures that sensitive data such as:

- Student records
- Financial transactions
- Research data

remains protected in transit and cannot be intercepted or modified.

7.3.4 Secure Remote Administration

All network devices (routers, switches, firewalls) are managed using secure protocols only:

- SSH (no Telnet)
- SNMPv3 (encrypted monitoring)
- Role-based admin access

This prevents credential theft and unauthorized configuration changes.

Function	Technology
Segmentation	VLANs + L3 routing
Perimeter Security	NGFW
Internal Security	ACLs + Internal Firewalls
Remote Access	IPSec/SSL VPN
Threat Detection	IDS
Threat Prevention	IPS
Encryption	TLS/HTTPS/SSH

Table 24. Security Technology Stack

7.4 Security Design Justification

The proposed security architecture is designed to meet the university's requirements for scalability, reliability, and strong data protection across all campuses.

7.4.1 Alignment with Requirements

This design directly supports Stage 1 requirements by ensuring:

- Secure access to LMS and academic platforms
- Protection of financial and student records
- Safe remote access for staff and researchers
- Isolation of guest and public traffic
- Secure inter-campus communication

7.4.2 Why This Design Works

This architecture is effective because it:

- Reduces attack surfaces through segmentation
- Prevents lateral movement using internal controls
- Protects external services using DMZ design
- Ensures continuous monitoring via IDS/IPS
- Enforces strict access control via Zero Trust principles

7.4.3 Scalability Consideration

The design is scalable because:

- New campuses can be added by extended VLAN and subnet structures
- Security policies are centrally managed
- Firewall rules can be replicated across sites
- VPN infrastructure supports distributed expansion

8. Individual Reflection

My Contribution

In Assignment 2, I was responsible for designing the network routing architecture and performance architecture for the Greenfield University multi-campus network. My work translated the requirements identified in Stage 1 into a comprehensive logical design that ensures reliable, high-performance connectivity across all three campuses.

Written deliverables I completed

- **Network Routing Architecture (Section 4):** Developed the complete routing strategy including OSPF selection justification over EIGRP and RIP, multi-area OSPF design with route summarization, BGP configuration for dual ISP failover, routing policies, and redundancy strategies with ECMP and BFD.
- **Network Performance Architecture (Section 5):** Designed the full performance framework including scalability strategy, DiffServ QoS architecture with six traffic classes and DSCP markings, LLQ and WRED implementation, policing and shaping policies, and formal Service Level Agreements (SLAs) with measurable targets.
- **Technical Justifications:** Provided comprehensive rationale for all design decisions, linking each choice back to Stage 1 challenges (c1, C3, C4, C6) and stakeholder requirements, ensuring traceability from requirements to implementation.

My Peers' Contribution

Maria served as the Addressing and Design Integration Lead, completing the Introduction (Section 1) and the Network Addressing Architecture (Section 3). She established the project overview, design objectives, summary of Stage 1 requirements, and design assumptions, providing the foundation and context for my routing and performance decisions. Her comprehensive IPv4/IPv6 addressing plan with VLSM allocation and VLAN tables directly informed my OSPF area design, as the hierarchical addressing enables efficient route summarization.

Saha served as the Network Diagram and Topology Design Lead, completing the Logical Network Diagram (Section 2). Her visual representation of the hierarchical core-distribution-access topology, VLAN segmentation, and key network devices provided the structural framework for my routing architecture. The topology she illustrated, showing Campus A as the core with Campuses B and C as satellite locations, directly influenced my multi-area OSPF design decisions.

Natalee served as the Security and Management Architectural Lead, completing the Network Management Architecture (Section 6) and Security Architecture (Section 7). Her security architecture, including firewall placement, VLAN segmentation, and VPN design, influenced my routing decisions regarding route redistribution boundaries at security zones. Her network management framework with SNMP, NetFlow, and IP SLA provided the monitoring infrastructure needed to verify the QoS and performance SLAs I defined.

Challenges Faced and Resolved

The main challenge I encountered was ensuring my OSPF area design and QoS bandwidth allocations aligned with the addressing scheme and topology developed by my peers. My initial OSPF design treated the entire network as a single area, which would have caused excessive LSA flooding and routing table sizes as the network expanded. I resolved this by redesigning to a multi-area hierarchy with Campus B and C as stub areas, referencing the /16 campus prefixes from Maria's addressing plan to define clear area boundaries and Saha's topology diagram to understand the physical relationships between campuses.

Another significant challenge was balancing the QoS bandwidth allocations. My first draft allocated 40% to EF traffic (video conferencing), which would have starved other critical applications during peak usage. After reviewing the application requirements Maria documented, I adjusted the allocations to 30% EF, 35% AF41, 15% AF21, 10% BE and 5% each for Network Control and Scavenger, ensuring all priority classes received adequate bandwidth while preventing any single class from monopolizing link capacity.

The final challenge was incorporating BFD with OSPF for sub-second convergence. Initially, I assumed OSPF's default timers would suffice, but the Stage 1 requirement for

<5 second failover demanded faster detection. I researched BFD integration and designed it into the routing architecture, ensuring the network meets the availability requirements defined in the project objectives.

What I Learned

I learned the routing and performance design cannot be completed in isolation, they are deeply interdependent with addressing, topology, and security architectures. The hierarchical addressing scheme directly enables route summarization, which reduces OSPF LSA flooding and improves convergence. The physical topology determines OSPF area boundaries and summarization points. Security zone boundaries affect where route redistribution can occur. This reinforced the importance of cross-functional collaboration within the team.

I gained practical skills in designing QoS architecture that balances competing requirements. I discovered that bandwidth allocation is not simply about giving priority to important traffic but also requires protecting less important traffic from starvation and preventing priority queues from consuming the entire link. The 30% LLQ cap is a critical safety mechanism that I learned from Cisco best practice documentation.

I also learned the importance of SLA definition and monitoring. Designing a network is incomplete without defining how its performance will be measured and verified. The IP SLA and NetFlow monitoring framework, which Natalee documented in Section 6, ensures that the performance commitments made in the design can be validated in operation, providing accountability to stakeholders.

Finally, I learned that routing protocol selection requires careful consideration of operational constraints. While EIGRP may offer faster convergence in some scenarios, OSPF's open-standard nature aligns with the university's requirement for vendor flexibility and long-term cost control. This reinforced the lesson that technical decisions must balance performance with business and operational requirements.

References

[1]

J. D. McCabe, *Network Analysis, Architecture, and Design*, 3rd ed. Amsterdam: Morgan Kaufmann Publishers, 2007.

[2]

“CCDE Study Guide: Enterprise Campus Architecture Design > Enterprise Campus: Hierarchical Design Models | Cisco Press,” *www.ciscopress.com*. https://www.ciscopress.com/articles/article.asp?p=2448489

[3]

A. Manzoor, M. Hussain, and S. Mehrban, “Performance Analysis and Route Optimization: Redistribution between EIGRP, OSPF & BGP Routing Protocols,” *Computer Standards & Interfaces*, vol. 68, p. 103391, Feb. 2020, doi: https://doi.org/10.1016/j.csi.2019.103391.

[4]

Cisco, “Cisco IOS XR Deployment Best Practices for OSPF/IS-IS and BGP Routing,” *Cisco*. https://www.cisco.com/c/en/us/support/docs/ios-nx-os-software/ios-xr-software/IOS-XR-Best-Practices/IOSXR-Deployment-BestPractices.html

[5]

A. Gupta, “Design BGP and SDN From Scratch,” *Ucsb.edu*, Apr. 09, 2026. https://sites.cs.ucsb.edu/~arpitgupta/cs176c/spring26/assets/slides/week02_lecture04_predictions.html#/title-slide (accessed Jun. 19, 2026).

[6]

Data Centre and Cloud, “Data Centre and Cloud,” *Cityu.edu.hk*, 2026. https://www.cityu.edu.hk/csc/deptweb/about/campusbackbone-tc.htm (accessed Jun. 19, 2026).

[7]

“IPQoS Architecture and the diffserv Model (IPQoS Administration Guide),” *Oracle.com*, 2026. https://docs.oracle.com/cd/E19683-01/816-4094/6mab39uur/index.html (accessed Jun. 19, 2026).

[8]

“Chapter 6 IPQoS in Depth (Reference) (IPQoS Administration Guide),” *Oracle.com*, 2026. https://docs.oracle.com/cd/E19683-01/816-4094/6mab39uup/index.html (accessed Jun. 19, 2026).

[9]

C. Bouras, A. Karaliotas, M. Oikonomakos, M. Paraskevas, D. Primpas, and C. Sintoris, “QoS issues in the Research and Academic Networks: The case of GRNET,” *Joint International Conference on Autonomic and Autonomous Systems and International Conference on Networking and Services - (icas-isns’05)*, pp. 60–60, 2005, doi: https://doi.org/10.1109/icas-icns.2005.80.